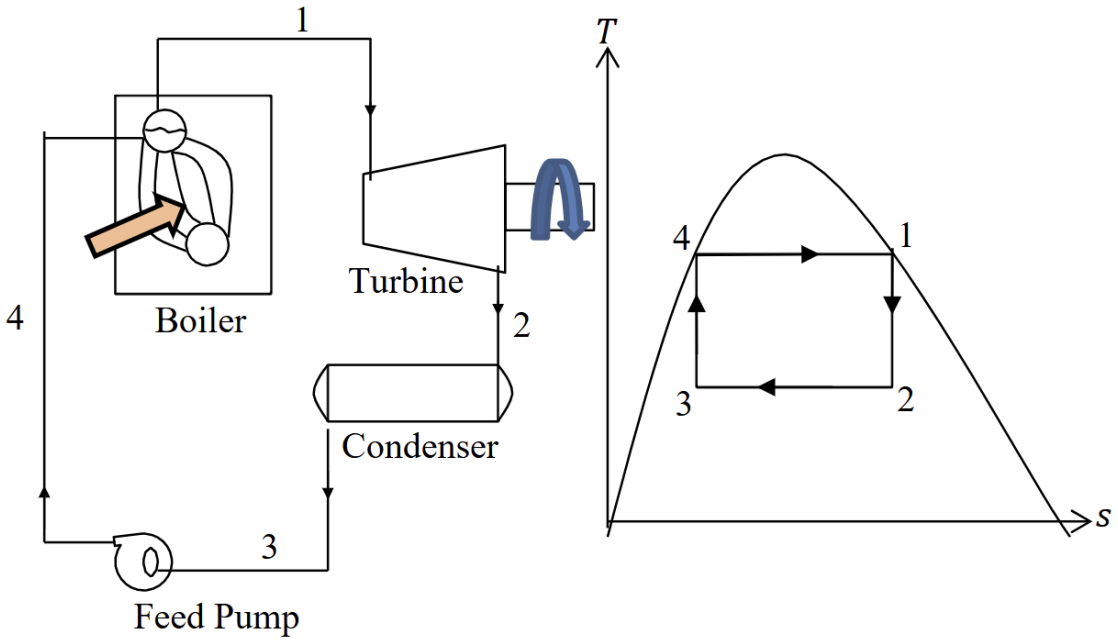
Vapour Powered Cycles

Vapour Powered cycles are similar to gas powered cycles but instead use steam as the working fluid.

The Carnot Vapour Cycle

Like the Carnot Gas Cycle, heat is transferred isothermally.



1 → 2: Isentropic Expansion

Steam is expanded adiabatically as it does work against the turbine, causing a work output.

2 → 3: Isothermal heat rejection

Steam is cooled in the condenser isothermally, while still remaining as wet steam.

3 → 4: Isentropic Compression

Steam is compressed adiabatically for it to reach the boiler pressure.

4 → 1: Isothermal heat supply

Steam is heated isothermally in the boiler.

Similar to the Carnot Gas Power Cycle, the efficiency is dependent on the temperatures of the hot and cold reservoirs, being:

—

$$\eta = 1 - \frac{T_2}{T_1}$$

where T2 is the temperature of the cold reservoir, and T1 is the temperature of the hot reservoir.

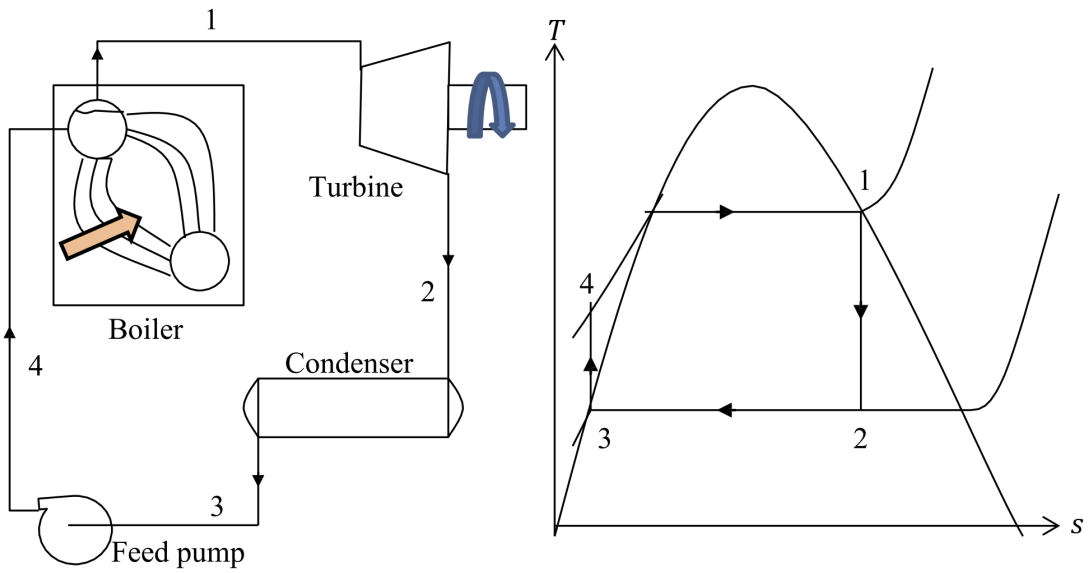
Like the Carnot Gas Power Cycle, this process is very idealised, and is impractical to implement, as:

The feed pump needs to be very large to be able to compress the steam to the boiler pressure. Hence, the work ratio for the Carnot Vapour Cycle is very low as well.

Isothermally supplying and rejecting heat from the system takes a long time, and is impractical to implement.

The Rankine Cycle

The Rankine Cycle is synonymous with the Diesel Cycle, in that heat is supplied and rejected at a constant pressure.



1 → 2: Isentropic Expansion

Steam is expanded adiabatically as it does work against the turbine, causing a work output.

2 → 3: Isobaric heat rejection

Steam is cooled in the condenser at constant pressure, reaching saturated water.

3 → 4: Isentropic Compression

The saturated water is then compressed to be pressurised for the boiler.

4 → 1: Isothermal heat supply

The liquid is heated into steam at a constant pressure for it to do work in the turbine.

Recall the simplified SFEE equations for (adiabatic) turbines/pumps and boilers/condensers:

$$w = h_1 - h_2$$

For (adiabatic) turbines and pumps:

for a process 1 → 2.

For boilers and condensers:

$$q = h_2 - h_1$$

for a process 1 → 2.

Hence, the work ratio for the Rankine Cycle is

$$\begin{aligned} \text{Work ratio} &= \frac{W_i - W_o}{W_i} \\ &= \frac{(h_1 - h_2) - (h_4 - h_3)}{h_1 - h_2} \end{aligned}$$

and the thermal efficiency can be found as

$$\begin{aligned} \eta &= \frac{\sum W}{Q_s} \\ &= \frac{(h_1 - h_2) - (h_4 - h_3)}{h_1 - h_4} \end{aligned}$$

However, the work due to the feed pump, $h_4 - h_3$, is usually small compared to the turbine work, and when neglected,

$$\eta = \frac{h_1 - h_2}{h_1 - h_4}$$

We can also quantify how much steam is required for 1kWh of work, using the specific steam consumption (SSC):

$$SSC = \frac{3600}{w} kg/kWh, \text{where } w \text{ is the specific net work done}$$

Another thing to note is that the isentropic expansion is not reversible in reality, and the expansion would be from $1 \rightarrow 2'$

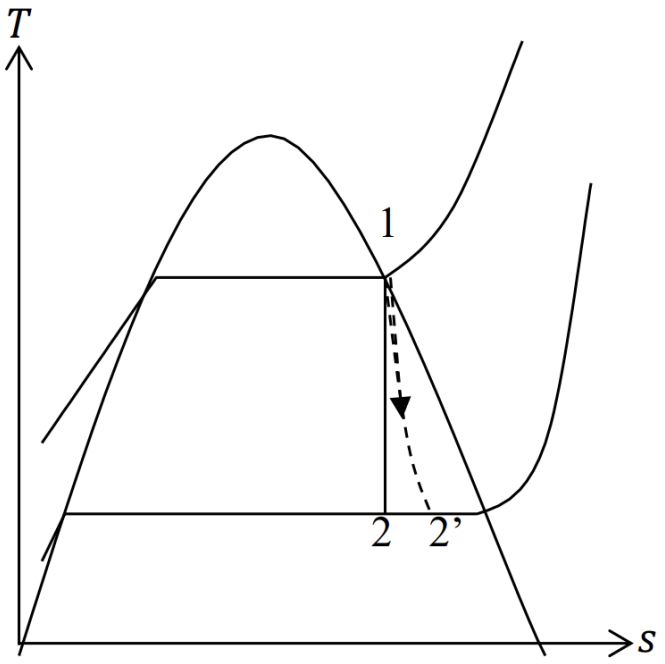


Figure 4-5: h_2, and h_2 of turbine

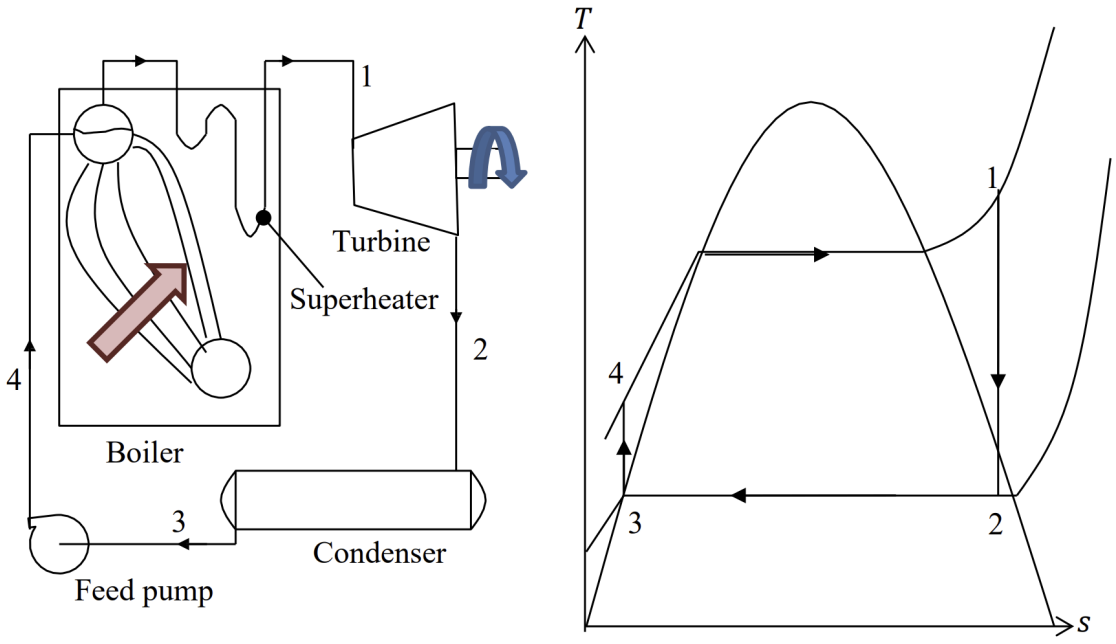
hence, we can compare the work done between the ideal and actual expansion processes to form the isentropic efficiency:

$$\eta_i = \frac{h_1 - h_{2'}}{h_1 - h_2}$$

Typically, the isentropic efficiency is given, hence it is more useful to use this formula to find $h_{2'}$

One way to improve the Rankine Cycle is by superheating the steam before sending it to the turbine.

The Rankine Cycle with Superheat



The benefits include:

higher thermal efficiency

reduced SSC

higher dryness fraction after turbine

lack of wet steam at turbine, which could corrode the blades.

Again, as this is still a Rankine cycle, the efficiency is

$$\begin{aligned}\text{Work ratio} &= \frac{W_i - W_o}{W_i} \\ &= \frac{(h_1 - h_2) - (h_4 - h_3)}{h_1 - h_2}\end{aligned}$$

or when neglecting the work due to the pump:

$$\eta = \frac{h_1 - h_2}{h_1 - h_4}$$

The Rankine Cycle with Reheat

This is similar to the dual combustion cycle, in that work is done in two steps.

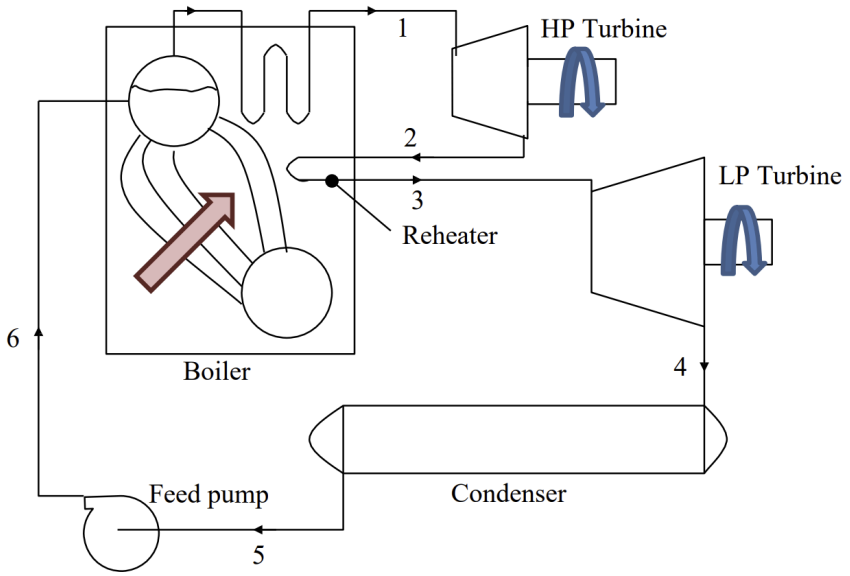
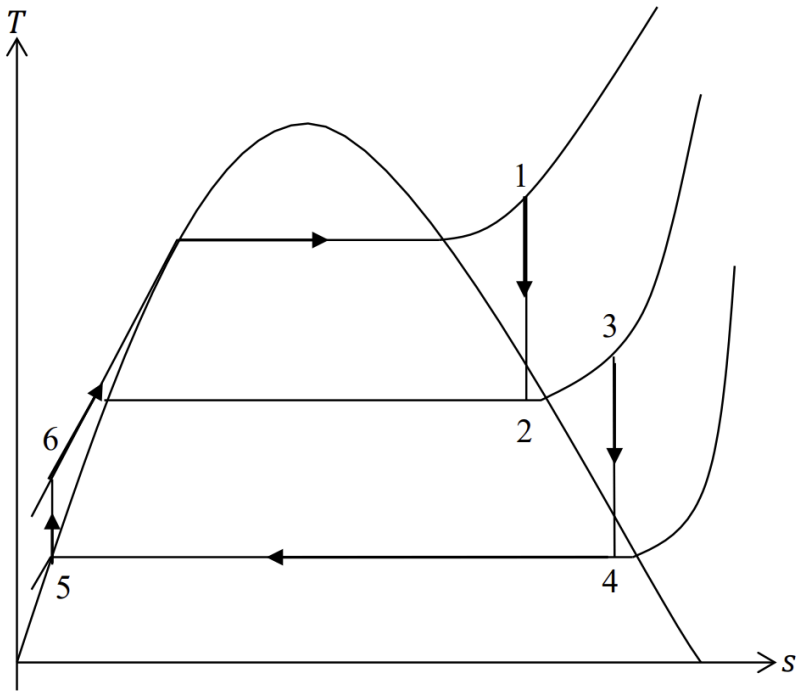


Figure 4-7: Rankine cycle with reheat



1 → 2: Isentropic expansion at high pressure

Superheated steam leaves the boiler and expands adiabatically at high pressure, to reach an intermediate pressure

2 → 3: Isobaric reheating

The wet steam is then rerouted into a reheater, to be superheated at constant pressure.

3 → 4: Isentropic expansion at low pressure

The reheated steam then expands at a lower pressure, reaching the pressure needed to enter the condenser.

4 → 5: Isobaric heat rejection

5 → 6: Isentropic compression

6 → 1: Isobaric heat supply to superheat

Typically for this cycle, the work due to the pump is negligible, and can be ignored.

Hence,

$$\begin{gathered}\Sigma w = w_{HP} + w_{LP} \\ = (h_1 - h_2) + (h_3 - h_4) \\ q_s = \text{heat supplied in boiler} + \text{heat supplied in reheater} \\ = (h_1 - h_6) + (h_3 - h_2) \\ \therefore \eta = \frac{\Sigma w}{q_s} = \frac{(h_1 - h_2) + (h_3 - h_4)}{(h_1 - h_6) + (h_3 - h_2)}\end{gathered}$$

And if the two turbines do not undergo ideal isentropic expansion,

$$\eta_{HP} = \frac{h_1 - h_{2'}}{h_1 - h_2},\ \eta_{LP} = \frac{h_3 - h_{4'}}{h_3 - h_4}$$

hence causing

$$\begin{gathered}\Sigma w = (h_1 - h_{2'}) + (h_3 - h_{4'}) \\ q_s = (h_1 - h_6) + (h_3 - h_{2'}) \\ \therefore \eta = \frac{\Sigma w}{q_s} = \frac{(h_1 - h_{2'}) + (h_3 - h_{4'})}{(h_1 - h_6) + (h_3 - h_{2'})}\end{gathered}$$